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Title: Early evidence for domestic chickens (*Gallus gallus domesticus*) in the Horn of Africa

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Abstract

Domestic chickens (*Gallus gallus domesticus* L. 1758) are one of the most valued farm animals in the world today. Chickens are widespread and economically and socially significant in Africa. Despite their importance little is known about the nature of their introduction and subsequent integration into African economies. One reason for this is the morphological similarity of domestic chickens to wild galliform birds in Africa such as Guinea fowl and francolin. Here we present direct dates and morphological evidence for domestic chickens recovered from Mezber, a pre-Aksumite (>800–450 BCE) rural farming settlement in northern Ethiopia. Key morphological markers differentiated these domestic chickens from francolins. The Mezber direct chicken element AMS dates of cal 820-595 BCE and indirect/charcoal AMS dates of cal 921-801 BCE constitute the earliest osteological evidence for chickens in Africa. Chicken bones in the domestic food waste of an early rural settlement at Mezber and their presence in later Aksumite urban contexts shows that chickens were integrated into diverse Ethiopian highland settings. The Mezber specimens predate the earliest known Egyptian chickens by at least 550 years and draw attention to early exotic faunal exchanges in the Horn of Africa during the early first millennium BCE. These findings support previous archaeological, genetics and linguistic data that suggests maritime exchange networks with South Arabia through ports along the African Red Sea coast constitute one possible early route of introduction of chickens to Africa.

Key words:

Chickens; Earliest osteological evidence; Direct AMS date; Red Sea; Horn of Africa; Routes of transport

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Introduction and background to chickens in Africa

Chickens (*Gallus gallus domesticus* L.) are a key domestic species globally and a mainstay of rural households in Africa and many regions of the world. Their widespread modern distribution reflects ancient processes of domestication and cultural contact. Associated with female domains today, chickens provide a small but reliable quantity of high protein food—in the form of meat and eggs—with relatively little tending (Mwalusanya *et al*., 2002; Sonaiya *et al*., 1999). Recent genetic research suggests that, by 6000 BCE, chickens had been domesticated multiple times within the range of the red jungle fowl *Gallus gallus*, between southern China and northern India (West & Zhou, 1989; Elliott *et al.*, 1994: 278–554; Blench, 2006; Liu *et al.*, 2006; Sawai *et al.*, 2010; Gifford-Gonzalez & Hanotte, 2011; Peters *et al.*, 2015). Small-scale interactions and trade along ancient land and sea routes ultimately led to chickens spreading across Asia and most regions of the world, including Africa (Perry-Gal *et al*., 2015). The dietary significance of chickens (Gifford-Gonzalez & Hanotte, 2011; Perry-Gal *et al*., 2015) and their role in cockfighting (Sykes 2012) are thought to have been factors in the spread of chickens globally. Evidence for early economic exploitation of the chicken outside the initial area of domestication is best known from the site of Maresha in the southern Levant dating to the fourth and second century BCE (Perry-Gal *et al*., 2015). Bio-recovery, however, has been poor on many sites and there is little in the way of faunal data bearing on the initial domestication and timing and routes of spread of chickens. Placing the origin and spread of chickens geographically and temporally is significant for social agricultural histories and for understanding present-day biodiversity.

In Africa today, chickens fulfill important economic and ritual roles in rural societies (Kitalyi, 1999; Sonaiya *et al.*, 1999; Blench, 2006). Ancient root words for chickens in African language phyla suggest that domestic chickens were introduced at least three times

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via two separate routes (Johnston, 1886; Blench, 1994; Williamson, 2000): from North Africa through the Sahara to West Africa; and from the East African coast to Central Africa (Williamson, 2000). Molecular genetic studies of modern African village chickens support this emphasis on multiple introductions and indicate the presence of fighting breeds of both Indonesian chickens and chickens of Indian and Southeast Asian origin (Blench, 2006; Muchadeyi *et al.*, 2008; Adebambo *et al.*, 2010; Gifford-Gonzalez & Hanotte, 2011; Mwacharo *et al.*, 2011). These studies tie the African Haplogroup E to Ethiopia and Sudan, and suggest that the origin of these chickens was Southern China (Gifford-Gonzalez & Hanotte, 2011; Mwacharo *et al.*, 2011, 2013). These studies also show that the biodiversity of present day African village chickens is higher than that of commercial breeds. Mwacharo *et al.* (2013) suggest that this might be a result of multiple reintroductions. However, there is surprisingly little direct archaeological evidence of the timing, geographical routes or social contexts of the spread of chickens to Africa (MacDonald, 1992; Boivin *et al.*, 2013).

Archaeological evidence has suggested that chickens were first introduced to Africa through North Africa, Egypt and the Nile Valley between the mid-second and late first millennium BCE. The earliest osteological evidence of chickens in Africa dates to the late first millennium BCE, from the Saite levels at Buto, Egypt—approximately 685-525 BCE (von den Driesch, 1997; Redding, 2015). Early chicken remains were also recovered from Phase B at Carthage, Tunisia dating to the fourth and fifth centuries BCE (Nobis, 1999). Chickens are also known from Tell Maskhuta, a Ptolemaic and early Roman site in Egypt, 332 BCE–200 CE (Boessneck, 1984), and from the site of Qasr Ibrim in Nubia, c. 450–550 CE (MacDonald & Edwards, 1993) as well as the urban site of Jarma in Libya dating to 100-200 CE (Mattingly, 2013).

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Interestingly, the earliest known representations of chickens in Africa are older than the faunal evidence, dating to the mid–second millennium BCE at ancient Egyptian and Meroitic period sites. An Egyptian ostracon depicting a rooster dates to the New Kingdom, c.1425-1123 BCE (Carter, 1923; Houlihan and Goodman; 1986; MacDonald & Edwards, 1993), whereas domestic chickens illustrated on Meroitic pottery and an ivory plaque from the tomb of Iretirou at Nuri date to c. 650 BCE (Dunham, 1955; Lobban, 2004: 146). These depictions, together with a lack of associated osteological evidence, have been interpreted as an indication that chickens were exotic species—imported for cock fighting or ritually important, but not fully incorporated into the domestic economy of ancient Egypt and Meroë for another thousand years (MacDonald & Edwards, 1993; Blench & MacDonald, 2000). It is also possible, that early artistic depictions of chickens were a reflection of exotic fowls observed by the artist during travel to Asia rather than local species.

In West Africa, osteological evidence for chickens is later than that found in North Africa. Chicken remains dating to the early first millennium CE to the last thousand years are known from Kirikongo and Jenne-jeno (Connah, 1981; MacDonald, 1992; MacDonald & Edwards, 1993; MacDonald & MacDonald, 2000; Dueppen, 2011). The ritual role of chickens is clearly indicated as early as c. 100 CE at Kirikongo, an Iron Age site in West Africa (Dueppen, 2011).

Recently, studies of trade in the Indian Ocean have drawn attention to Eastern Africa as an early avenue of introduction of Indian and Southeast Asian chickens (Fuller *et al*., 2011; MacDonald, 1992; Boivin *et al.*, 2013; Boivin *et al.*, 2014). Chickens have been recovered from levels dating to c. 500–1200 CE at both coastal and inland sites such as Unguja Ukuu, Akameru and Cyinkomane, and from sites in the Lamu Archipelago (Horton & Mudida,

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1996; Van Neer, 1990). Significantly earlier dates for chickens at Machaga Cave, c. 3000-2000 BCE (Chami, 2001; 2007) are contested (Boivin *et al*., 2013; Mwacharo *et al.*, 2013; Dueppen, 2011). Chickens date to similar periods in Central and Southern Africa (Table 1 and Figure 1; MacDonald, 1992; Boivin *et al.*, 2013; Juma, 2004). None of the aforementioned chicken remains has been directly dated, but, together, their presence suggests that chickens were widely distributed in Africa at least by 1200 CE.

In addition to regional gaps in research coverage, chronological and methodological concerns have played a role in limiting our understanding of the timing, routes and process by which chickens spread in Africa (MacDonald, 1992). It is currently unclear whether the chronological gap between chicken remains at North African sites dating to the late first millennium BCE and those from the rest of Africa dating to the second millennium CE reflects patterns of introduction or artifacts of method. MacDonald (1992: 303) cites the recovery bias against small fauna, the low density of bird remains in sites and the co-occurrence of osteologically similar African wild fowls (including guineafowl, *Numida meleagris* L. and francolin, *Francolinus* sp.) with domestic chickens as significant factors limiting our understanding of the dispersal of chickens in Africa. These problems are not unique to Africa, however. In general, the osteological study of bird bones has focused on a small number of species, globally, and has tended to be regionally focused (Bocheński & Tomek, 2009; Serjeantson, 2009; Perry-Gal *et al*., 2015; Tomek & Bocheński, 2009; Peters *et al.*, 2015).

We report here on the identification, archaeological context and direct AMS (Accelerator Mass Spectrometry) dating of early chicken bones excavated from pre-Aksumite (>850-400 BCE) deposits at the archaeological site of Mezber, located in the Eastern Tigrai region of the

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northern Ethiopian highlands (Figure 2). These direct dates were the first to be obtained from chickens in Africa and provide new archaeological information on the dispersal patterns of domestic chickens, complementing genetic data. Our findings demonstrate that chickens played a subsistence role in pre-Aksumite society, and draw attention to the role of exchange and trade across the Red Sea in the early introduction of chickens to Africa through pre-Aksumite networks in the Horn of Africa.

Mezber: Archaeological context

Mezber is a pre-Aksumite rural settlement in the Gulo Makeda administrative zone of Eastern Tigrai (Figure 2). The site lies approximately 78 km from Yeha, a pre-Aksumite urban settlement, and 34 km from Aksum, the capital of the Aksumite state. It lies on a slightly raised hillside situated in a valley bottom and covers approximately 0.83 ha. The site was discovered in 2005 during archaeological surveys completed by the Eastern Tigrai Archaeological Project (ETAP) in northeastern Tigrai (D'Andrea *et al.*, 2008; Harrower & D'Andrea, 2014). Mapping and excavations were carried out by ETAP in 2007–2009 and 2011–2012. Mezber is one of only a few known pre-Aksumite sites that have been undisturbed by later settlement. Excavations have revealed an extended occupational sequence spanning the mid-second to the end of the first millennium BCE. Four phases are evident, based on ceramics, stratigraphy and radiocarbon dating, and at least two of these were building construction phases. The earliest occupation of the site is during an Initial phase (c. cal 1600-900 BCE) followed by Early (c. cal 850-750 BCE), Middle (c. cal 600–400 BCE) and Late (c. cal 400 BCE–1BCE/CE) pre-Aksumite phases. The site has produced large-scale domestic architecture, suggesting the presence of wealthy or elite individuals in the Early Phase, as well as evidence of specialized craft production in the Middle Phase (D'Andrea & Welton, in prep.).

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Zooarchaeological data

Excavations at Mezber were conducted with specific attention to the retrieval of zooarchaeological and archaeobotanical remains. All excavated soil beneath the plough zone was sieved through 5 mm mesh and, where possible, faunal specimens were recovered in hand excavations prior to screening. Systematic flotation was also conducted. The chicken bones reported here were recovered from excavated, rather than floated, samples. Zooarchaeological identifications were made using ETAP reference collections and recorded according to African zooarchaeological protocols (Gifford & Crader, 1977; Walker, 1985; Marshall, 1990; MacDonald, 1992; Peters, 1995).

The ongoing study of the Mezber zooarchaeological assemblage is providing an unprecedented view of livestock economies and animal husbandry practices during the pre-Aksumite period in the Horn of Africa. The Mezber zooarchaeological assemblage analyzed to date amounts to a number of identified specimens (NISP) of 5,192. Well preserved remains from pre-Aksumite contexts include cattle, which may be either African *Bos taurus*, Asian *Bos indicus* or hybrids, sheep or goat (Caprini), chicken, donkey (*Equus asinus africanus* L.) and dog (*Canis familiaris* L.). Wild species recovered include francolin (*Francolinus*), hare (*Lepus*), gazelle (*Gazella*), hyrax (Procaviidae) and rodents. We focus here on the chicken remains. Stable isotopic studies of several Mezber pre-Aksumite species— including chickens—have been published (D'Andrea *et al.*, 2011), but the identification, context and direct AMS dates of chicken are reported here for the first time. The avian component makes up 3% of the assemblage and includes 42 domesticated chicken (*Gallus gallus domesticus*), 5 cf. *G. gallus*, 14 Francolinus sp., 8 cf. *Francolinus*, and 132 Aves indeterminate (Table 4). No small avian species were identified. Specimens of domestic chicken were distributed

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widely across the site in Fields A1, A2, C1 and E1 (Figure 3), in contexts dating to the Early, Middle and Late pre-Aksumite phases (cal 850 BCE–1 CE). Chicken remains were not, however, recovered from Initial phase deposits.

Chickens belong to the Galliform order of birds and part of the challenge of identifying chickens in Africa is distinguishing them from other Galliforms. The Galliformes belong to only one family in Africa: the Phasianidae, but the Phasianidae are divided into six subfamilies. Three of them—Numidinae, Pavoninae and Gallinae—occur in Africa (Urban *et al*., 1986; Del Hoyo, *et al*., 1994; Elliot *et al*., 1994).

The secure identification of the 42 domestic chicken elements was based on size and morphology (Bocheński & Tomek, 2009; Serjeantson, 2009). Because bird bones are rarely found in complete form, we examined and measured separately the distal and proximal ends of the morphological markers of bird bones (Serjeantson, 2009). We employed MacDonald's (1992) criteria for differentiating chickens from African wild fowls, complemented by Bocheński and Tomek's (2009) approaches to separating domestic from wild bird species. We found MacDonald's morphological criteria for distinguishing guineafowl, francolin and chicken to be supported by modern comparative francolin specimens collected near the site and guineafowl studied at the History of Veterinary Medicine and Institute of Palaeoanatomy in Munich. Coracoid, scapula, radius and tarsometatarsus elements were especially useful in the differentiation of chickens from francolins, *Francolinus* sp. The intermetacarpal tuberosity of the carpometacarpus and the spur on the tarsometatarsus proved especially useful for differentiating chickens and guineafowl, Numidinae (Figure 5) (MacDonald 1992). We also used the osteological size ranges compiled and published by MacDonald (1992) that differentiate chickens from francolin and guineafowl on the basis of metrics. Of the 42

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chicken elements identified at Mezber, 16 complete long-bones and long-bone ends were measured (Table 2), while the remaining 26 were identified on morphological grounds (Table S1) (Bocheński & Tomek, 2009; Tomek & Bocheński, 2009; Serjeantson, 2009). Samples were first separated on the basis of size, francolins tend to be smaller than chickens. Bone modifications included fine cut marks identified on chicken bones such as the extremitas omalis of a coracoid and distal tibiotarsus on 0.9% of the assemblage. The locations of these marks were consistent with defleshing or disarticulation. Human tooth marks were identified on the shaft of a humerus, these are relatively shallow, distinctively spaced, and consistent with those described by Landt (2007) (Figure 6).

Direct dating using AMS was completed on two domestic chicken specimens: a tarsometatarsus (Figure 4a) from an Early Phase context and a carpometacarpus from a Middle Phase deposit (Figure 4b). Table 3 summarizes the direct AMS dates obtained on chicken elements and the associated charcoal dates from Mezber. The Early Phase tarsometatarsus (2590 ± 30 BP, cal 820-595 BCE; A1 L47 P106 #1116) was excavated in a deep sounding in the northern section of Field A1, Locus 47. Early Phase deposits in this section of the site were comprised of a sandy fill with numerous and diffuse ash-like lenses associated with the interior portion of a large building in the lower architectural phase. In general, the preservation of faunal bones in Locus 47 was excellent and the high concentration of livestock and other animal bones suggests the accumulation of animal processing or food discard. The Middle Phase carpometacarpus (2390 ± 40 BP, cal 747-389 BCE; C1 L17 P42 #1164) was recovered from a deep sounding in Field C1, Locus 17, beneath a structure resembling a kitchen with a hearth and many large grinding stones (Figure 3). Locus 17, Pail 42, which produced the AMS-dated carpometacarpus, consisted of Middle Phase fill below the kitchen floor (Figure 3). In addition, two chicken specimens were found

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in securely dated Early and Middle phase contexts. A tibiotarsus (A1 L61 P136 #1868) recovered in an ash lens, between the upper and lower architecture, associated with charcoal dated to 2700±40 BP, cal 921–801 BCE. A coracoid (A2 L26 P41 #1274) recovered from a living floor between the upper and lower architecture was associated with donkey unciform was dated to 2550±40 BP, cal 805–542 BCE. Early Phase ceramics were also recovered from the same locus (Tables 2 and 3).

Discussion

The Mezber chicken remains dating to 2590 ± 30 BP (cal 820-595BCE) are the earliest to have been directly dated in sub-Saharan Africa. Chicken remains from a securely dated building context also indicate that chickens were present in the Horn of Africa by 900 BCE. This pre-dates the earliest direct osteological evidence of chicken in Ptolemaic Egypt by more than 550 years.

Narratives of the introduction and spread of chickens to Africa have suggested patterns of northern or eastern introductions through African–Asian exchanges. The discovery of pre-Aksumite period chickens at Mezber emphasizes the significance of the Indian Ocean and Red Sea trade for agricultural exchange and the introduction of chickens to Eastern Africa.

These new data also demonstrate that chickens had an early dietary role in pre-Aksumite society in rural settings and possible elite contexts. Chickens are relatively abundant (Table 4) at Mezber. Food preparation and a role for chickens in pre-Aksumite cuisine is indicated by their occurrence in fill associated with a room interpreted as a kitchen and among discarded food waste with other livestock, as well as by traces of cut marks and human chewing on their bones. Elsewhere in Ethiopia, domestic chickens account for 1.3% of the

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assemblage NISP in food waste deposit in less elite areas of a later Aksumite (600 CE) urban context at the D site in Aksum (Cain, 2000), and 0.1% of the assemblage NISP in an elite settlement of an early (50–40 BCE) and post (800 CE) Aksumite urban context at the Ona Nagast site in Aksum (Chaix, 2013). The abundance of chickens at Mezber might be related to differences in chicken utilization between rural and urban areas as well as areas excavated and preservation and recovery biases. Ceramic figures of what could be domestic chickens are associated with funerary contexts and elite burials, dating to the Classic Aksumite period (CE 150–350; e.g., D'Andrea *et al.*, 2008). This evidence may hint at the existence of a ritual or other non-economic role of domestic fowl during Aksumite times. To date, however, the Mezber evidence documents the use of chickens in food rather than in ritual.

How did the chicken cross the Red Sea? Timing and routes of introduction

The case for the introduction of chickens to the Horn of Africa via Red Sea maritime trading routes is strongly supported by several sources of information. Rich textual and archaeological evidence documents Aksumite trade between coastal settlements on the western and eastern shores of the Red Sea (Fattovich, 1990; Fuller *et al.*, 2010). The Mezber dates suggest that chickens were part of the pre-Aksumite trade network. Adulis was an important pre-Aksumite trading port, which connected the Ethiopian and Eritrean highlands to the maritime trade route along the Red Sea and Gulf of Aden during the first millennium BCE (Casson, 2012). Architecture, inscriptions and artifacts in northern Ethiopia indicate extensive contact with South Arabia beginning in approximately 850 BCE (Fattovich, 1990; Wolf & Nowotnick, 2010; Gerlach, 2012; Phillipson, 2012). Overland routes connected Adulis with Aksum/Yeha, and one of these is believed to have passed near Mezber (D'Andrea *et al.*, 2008; Harrower & D'Andrea, 2014). Chickens could also have been introduced to the Horn of Africa from the Somali coast. However, Mezber's connection to

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Red Sea trade routes supports the theory of an early introduction of chickens to the Ethiopian highlands and the likelihood of chicken's introduction from this region, rather than the Indian Ocean.

The Mezber findings are also significant for our understanding of the cultural contexts of faunal exchanges and the agricultural systems in place during the pre-Aksumite period. The chickens at Mezber pre-date by a small margin those in South Arabia, where chicken remains from the site of Al-Hamid in Yemen date to 800–600 BCE (Vagedes, 2013: 174). Arabia was, in turn, tied to South Asian, Harappan and Near Eastern networks (Fuller, 2003; Boivin *et al*., 2014; Perry-Gal *et al*., 2015). Chicken remains from the Indian subcontinent date to the second millennium BCE and to the third millennium BCE (Perry-Gal *et al*., 2015). The presence of chickens on west coast Harappan sites may indicate that they were transported from there, by boat, to the Arabian Peninsula (Fuller, 2003; McIntosh, 2008). The location of Al-Hamid—close to the Red Sea in the transition zone between the desert and coastal settlements of Yeman—demonstrates possible routes for the early movement of chickens from the Arabian Peninsula to the Horn. These routes would have allowed chickens of South or East Asian heritage to reach the Horn prior to the Indian Ocean trade of the 6th century CE. The Al-Hamid dates also suggest a rapid introduction of chickens to the Horn. This may reflect the ease with which chickens can be transported, as well as more deeply embedded cultural issues (Harrower *et al.*, 2010). The genetic similarities of present day chicken lineages in the Arabian Peninsula to those from the Horn of Africa and Sudan also support the early role of regions like Bahrain and Oman in the dispersal of chicken from their place of origin in Asia to Africa (Al-Qamashoui, 2014). Their adoption may have been facilitated by earlier reliance of people of the Horn on similar wild birds. Wild galliform birds (Phasianidae) such as francolin were part of the food refuse in Mezber deposits pre-dating

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900 BCE. Chickens appearing in subsequent Early Phase deposits may fit a well-recognized food category and have been quickly adopted into pre-Aksumite cuisines.

Ancient introduction of domestic animals to new regions was not always successful, however, as documented by die-offs and repeat introductions of the earliest domestic animals to Cyprus (Vigne *et al.*, 2011), or the more than 500 year time lag after early herders introduced sheep and goat to northern Kenya 5000-4000 years ago before cattle became widespread (Gifford- Gonzalez, 2000). There has been little discussion of the success or failure of chicken introductions globally, but the early dates for chickens at Mezber, cal 820–595 BCE, combined with their presence in the Middle Phase cal 600–400 BCE, and in Aksumite contexts 40 BCE - 600 CE, demonstrate their long-term success in northern Ethiopia. This provides a setting for contextualizing the high genetic diversity of modern chickens in Eastern Africa with a more than 1000 year period that allows for multiple reintroductions and contributions from diverse chicken lineages from early Arabian and South Asian contexts and the later Swahili networks.

Several conclusions can be drawn when viewing the Mezber chicken evidence through a broader regional context. It is clear that domestic chickens have been present in the Horn since the early first millennium BCE and persisted as a domestic species into Aksumite and later periods. Although the role of chickens in the later pre-history of the Horn of Africa was primarily economic, the possibility of additional social or ritual roles for chickens should not be excluded at this point. At least one of the multiple introductions of chicken to Africa took place via maritime exchange networks with South Arabia through ports along the African Red Sea coast, perhaps involving pan-Asian or Indian Ocean contacts farther east (Gifford-

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Gonzalez & Hanotte, 2011). Finally, these data draw attention to exchange with Southwest Asia, and contextualize the present genetic diversity of chickens in East Africa and the Horn.

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Supporting Information

Supporting table (S1) shows osteological sample of twenty-six partial chicken elements studied from Mezber with information on key landmarks used to differentiate within galliforms.

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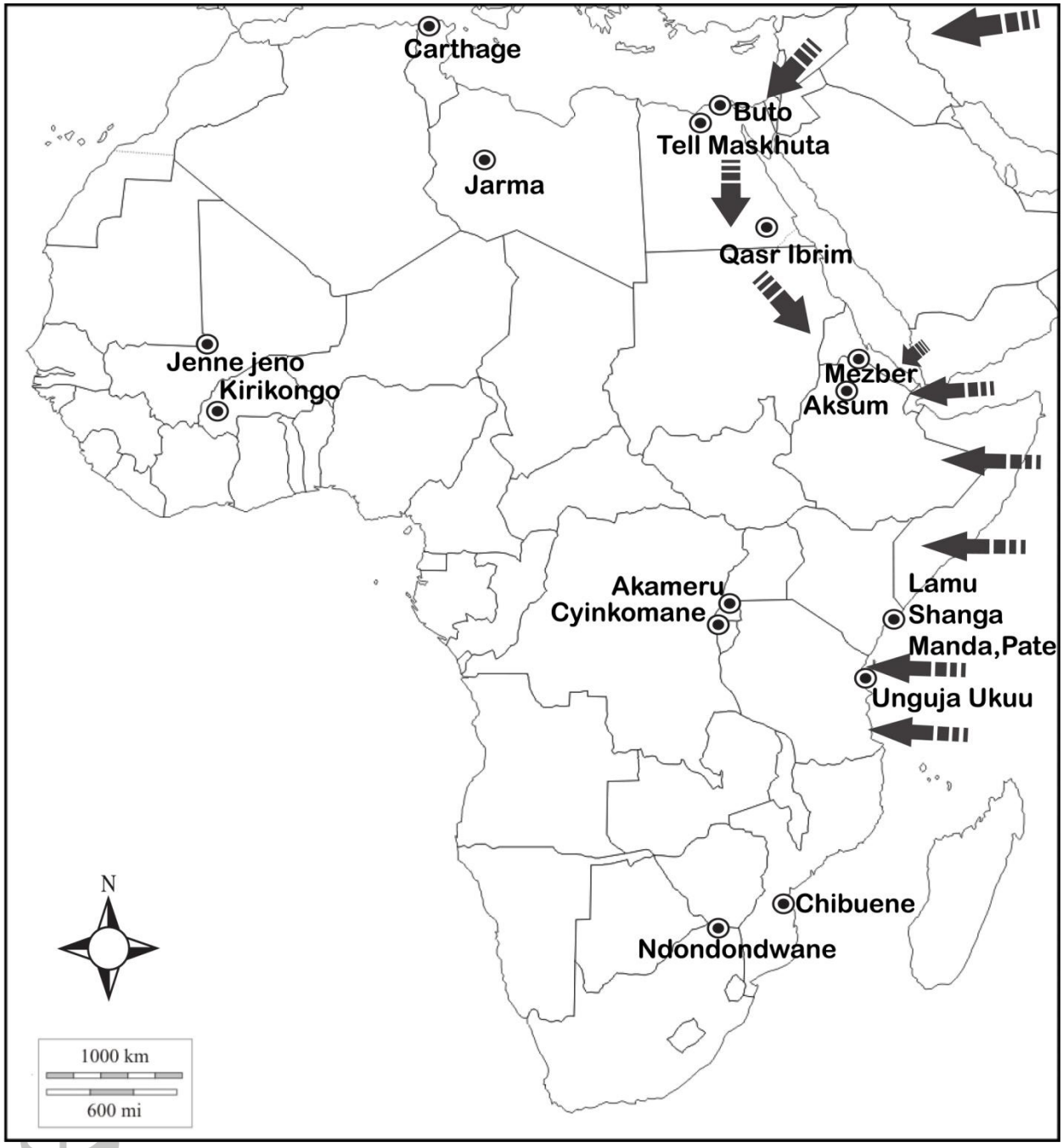


Figure 1. Map showing key archaeological sites mentioned in the text and possible routes of the introduction of chickens to Africa.

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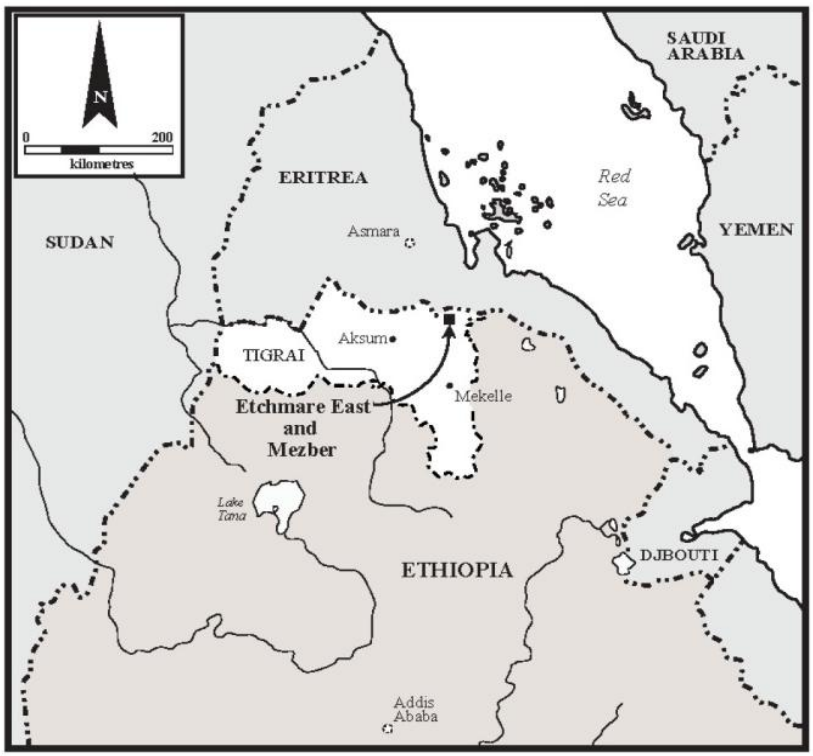


Figure 2. Location of the Mezber archaeological site, northern Ethiopia.

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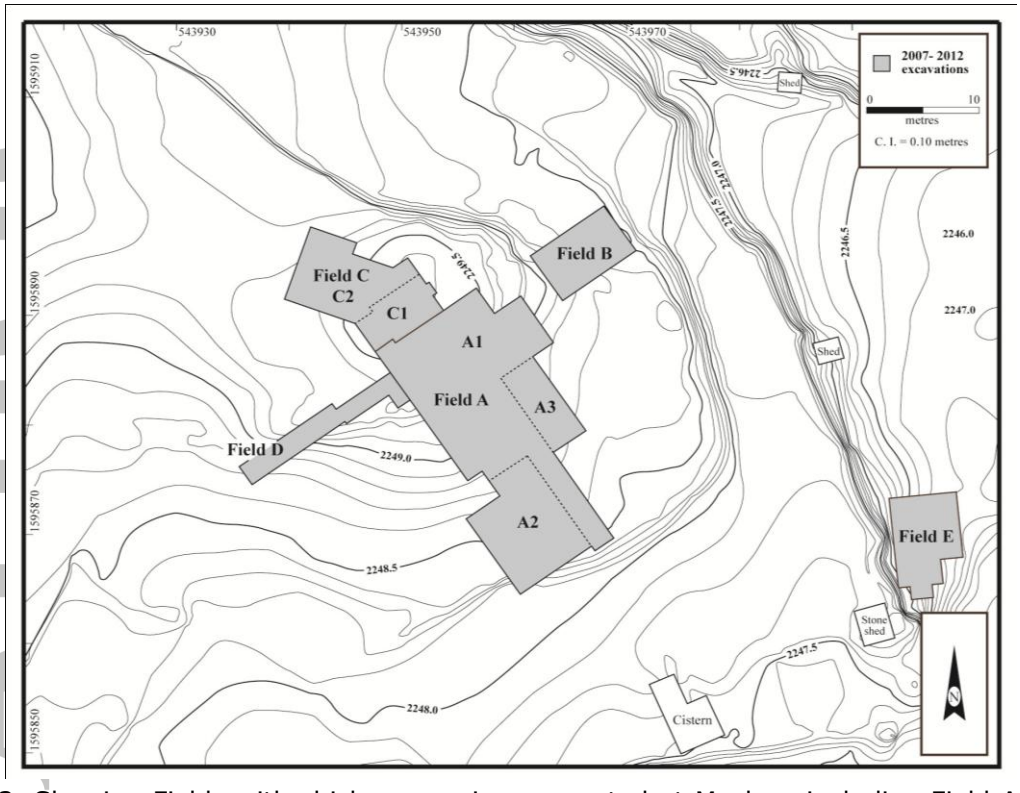


Figure 3. Showing Fields with chicken remains excavated at Mezber, including Field A, which yielded Early Phase chicken remains directly dated by AMS (Field A1 Locus 47 Pail 106 #1116). Map by Lynn Welton.

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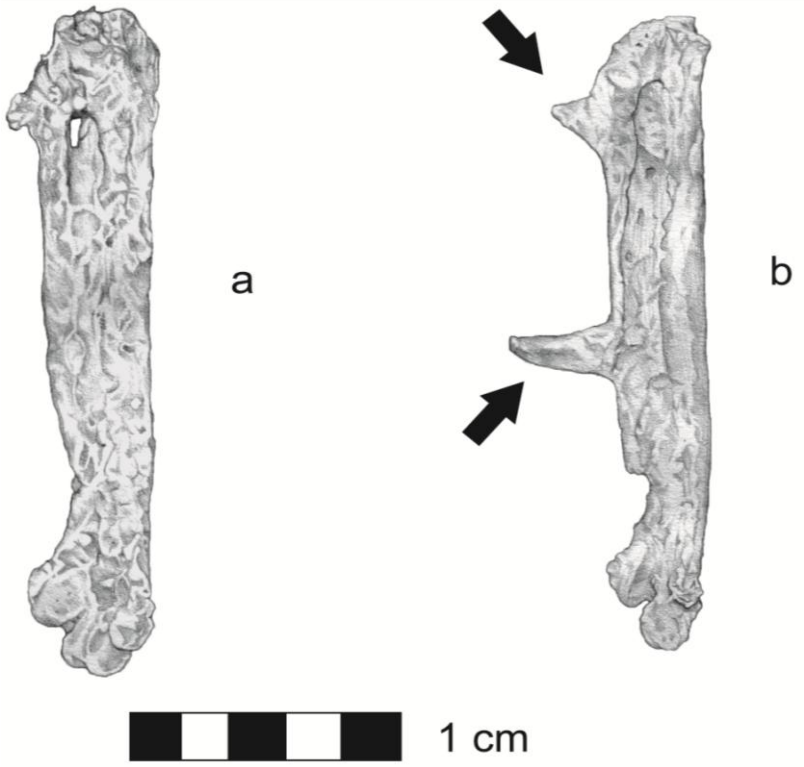


Figure 4. Chicken and francolin specimens from Mezber: A=tarsometatarsus, female domestic chicken with no spur, Field A1, Locus 47, Pail 106 #1116; B=tarsometatarsus, double-spurred, Francolinus, Field A1 Locus 47 Pail 95 #1056. Drawing by Esther Hamburger.

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Figure 5. Modern domestic chicken specimens from the Mezber region, showing gender distinctions and key landmarks used in identification. A=tarsometatarsus, female; B=spurred tarsometatarsus, male; C=carpometacarpus with intermetacarpal tuberosity. Drawing by Esther Hamburger.

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Figure 6. Some remains of domestic chickens from Mezber. a= distal tibiotarsus, cut mark (E1, Loc 4-Pail 6 #1663, Early Phase, 760-400 BCE), b= shaft of a humerus, human tooth mark (B2, Loc 4-Pail 13 #53, Middle Phase, 600-390 BCE).

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Table 1. Archaeological evidence for the earliest domestic chickens (*Gallus gallus*) in Africa

Site	Date	References
Mezber, Ethiopia	819–755 BCE	This article
Tell Maskhuta, Egypt	c. 332-200 BCE & 30-200 CE	Boessneck, 1986, MacDonald, 1992
Kirikongo, Burkina Faso	c. 100-1450 CE	Dueppen, 2011
Qasr Ibrim, Sudan	c. 450-550 CE	MacDonald and Edwards, 1993
Jenne-jeno, Mali	c. 450-850 CE & c. 850-1400 CE	MacDonald, 1992
Aksum, Ethiopia	c. 600 CE	Cain, 2000; Chaix, 2013
Akameru and Cyinkomane, Rwanda	c. 800-1100 CE	MacDonald, 1992, Van Neer, 1990
Manda, Lamu	Mid 9th-Early 13th CE	Chittick, 1984
Pate, Lamu	Mid 8th-Mid 13th CE	Wilson & Omar, 1997
Shanga, Lamu	Late 8th-15th CE	Horton & Mudida, 1996
Unguja Ukuu, Zanzibar	6th? –Mid 12th CE	Juma, 2004; Horton & Mudida, 2014
Chibuene, Mozambique	7th–10th/14th CE	Ekblom, 2004; Badenhorst *et al*., 2011

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Table 2. Osteological sample of sixteen chicken elements studied from Mezber with information on key landmarks and measurements used to differentiate within galliforms.

Ceramic Phase	Context*	Element***	Measurement (mm)	Key Landmark Features References (MacDonald 1992;Tomek and Bocheński 2009)	Dating	Isotopic Analysis (D'Andrea et al.
Early 850-750 cal BCE	Field A1 Locus 47 Pail 106 #1116	Tarsometatarus-GL Tarsometatarus-Bp Tarsometatarus-Bd	64.80 11.91 13.62	*One spur core, sulcus extensorius is shallow, no extra opening between trochlea metatarsi II and III, pecten hypotarsometatarsalis is missing, medial side of cotyla medialis clearly bulges* (1992:314; 2009:76-77)	2590±30 bp (820-595 cal BCE)**	
	Field A1 Locus 60 Pail 111 #1566	Coracoid-GL	40.90	*Facies articularis sternalis (midly concave)*(1992:311)		X
	Field A1 Locus 61 Pail 136 #1868	Tibiotarsus-GL Tibiotarsus-Bd	114.20 12.40	*Short and broad facies gastrocnemialis* (2009:65). *A slanting pons supratendineus* (2009:72)	Indirect charcoal date: 2700±40 bp (921-801 cal BCE)	X
	Field A1 Locus 62 Pail 120 #1609	Tarsometatarus-GL Tarsometatarus-Bp Tarsometatarus-Bd	82.21 13.32 14.43	*Sulcus extensorius is shallow, no extra opening between trochlea metatarsi II and III, pecten hypotarsometatarsalis is missing, medial side of cotyla medialis clearly bulges* (2009:76-77)		X
	Field A1 Locus 64 Pail 130 #1796	Carpometacarpus-GL Carpometacarpus-Bp	28.51 6.97	*Shallow trochlea carpalis notch* (2009:51). *Intermetacarpal tuberosity (highly developed)*(1992:313)		
	Field A2 Locus 26 Pail 41 #1274	Coracoid-GL	46.00	*Facies articularis sternalis (midly concave)*(1992:311)		
Early/Middle	Field A1 Locus 47 Pail 97 #1064	Tibiotarsus-Bd Tibiotarsus-Dd	9.90 8.94	*Short and broad facies gastrocnemialis* (2009:65). *A slanting pons supratendineus* (2009:72)	Indirect date: 2550±40 bp (806-542 cal BCE)	
	Field A1 Locus 71 Pail 151 #2081	Humerus, Distal-Bd	15.70	*Impressio musculi brachialis reaches to the ventral side of the shaft* (2009:34)		

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Table 2. (Continued)

Ceramic Phase	Context*	Element***	Measurement (mm)	*Key Landmark Features References (MacDonald 1992;Tomek and Bocheński 2009)*	Dating	Isotopic Analysis (D'Andrea et al.
Middle 600-400 cal BCE	Field A1 Locus 4 Pail 13 #81	Tibiotarsus-Bd	8.66	*Short and broad facies*		
		Tibiotarsus-Dd	7.22	*gastrocnemialis* (2009:65). *A slanting pons supratendineus* (2009:72)		
	Field A1 Locus 11 Pail 28 #58	Tibiotarsus-Bd	10.70	"		
		Tibiotarsus-Dd	10.09	"		
	Field A2 Locus 19 Pail 19 #884	Tarsometatarus-GL	60.00	*Lacking an extended hypotarsal ridge,*		
		Tarsometatarus-Bp	10.50	*single spur, sulcus extensorius is*		
		Tarsometatarus-Bd	10.00	*shallow, no extra opening between trochlea metatarsi II and III, pecten hypotarsometatarsalis is missing, medial side of cotyla medialis clearly bulges* (1992:314; 2009:76-77)		
	Field B2 Locus 3 Pail 7 #45	Tibiotarsus-Bd	10.46	*Short and broad facies gastrocnemialis* (2009:65). *A slanting pons supratendineus* (2009:72)		
	Field B2 Locus 4 Pail 12 #47	Tibiotarsus-Bd	12.88	*Short and broad facies*		
		Tibiotarsus-Dd	10.34	*gastrocnemialis* (2009:65). *A slanting pons supratendineus* (2009:72)		
	Field C1 Locus 17 Pail 42 #1164	Carpometacarpus-GL	56.00	*Intermetacarpal tuberosity (highly*	2390±40 bp (747-389 cal BCE)**	
		Carpometacarpus-Bp	16.00	*developed)*		
	Field E1 Locus 11 Pail 20 #2171	Tibiotarsus-Bd	9.63	*Short and broad facies*		
		Tibiotarsus-Dd	7.69	*gastrocnemialis* (2009:65). *A slanting pons supratendineus* (2009:72)		
	Field E1 Locus 14 Pail 17 #1982	Radius, distal-Bd	6.43	*Distal width of all domestic galliforms is larger than 5.5 mm* (2009:51)		

*"Field" is an excavation area, with individual squares numbered sequentially, e.g., A1, A2. "Locus" refers to a discreet depositional unit within a square, including walls and soil layers. "Pail" is a smaller sub-unit within a Locus. A serial number is a unique registration number assigned to separate categories of finds recovered from one pail, including artefacts, samples, and bone specimens.

** Direct AMS (Accelerator Mass Spectrometry) dates of domestic chicken bones.

***Measurements taken following Angela von den Driesch, 1976. GL= Greatest length; Bp= Breadth of the proximal end; Bd=Greatest breadth of the distal end.

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Table 3. AMS radiocarbon dates from Mezber, including chicken bones and charcoal from related loci. Ceramic phases at Mezber (Manzo in prep.) include: Initial (c. 1600-900 BCE); Early (c. 850-750 BCE); Middle (c. 600-400 BCE); and Late (c.400 BCE-CE 100).

Sample	Context	ID	Convent RC Age bp	OxCal 4.2, 95.4% prob cal BCE	Ceramic Phasing	Comment
Beta-298138	C1 L22 P41 #1158	Charcoal	2870 ± 30	1127-931	Initial	fill beneath lower architecture
Beta-378044	A1 L47 P106 #1116	Chicken	2590±30	820-595	Early	tarsometatarsus assoc. with lower architecture
Beta-266012	A1 L61 P115 #1589	Charcoal	2700±40	921-801	Early	fill between upper & lower architecture
Beta-266010	A2 L26 P40 #1268	Donkey	2550±40	805-542	Early	donkey unciform between upper & lower architecture
Beta-266009	C1 L17 P42 #1164	Chicken	2390±40	747-389	Middle	carpometacarpus in fill below surface
Beta-298137	C1 L20 P40 #1152	Charcoal	2370 ± 30	540-388	Middle	fill between upper and lower architecture
Beta-298134	C1 L3 P7 #372	Charcoal	2160±30	358-108	Late	fill associated with upper architecture
Beta-298135	C1 L8 P14 #400	Charcoal	2260±30	397-209	Late	floor/surface within upper possible kitchen building
Beta-298136	C1 L17 P34 #719	charcoal	2170±30	360-116	Late	Fill above surface
Beta-298133	C1 L3 P9 #37	Charcoal	2090 ± 30	193-42	Late	Fill associated with upper architecture

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Table 4. List of Aves remains from Mezber, Ethiopia

Context	A1	A2	A3	B2	C1	C2	D1	E1	
Domestic species:	NISP	NISP	NISP	NISP	NISP	NISP	NISP	NISP	Total
Chicken, *Gallus gallus*	16	4		10	1	1		10	42
cf. chicken, cf. *Gallus gallus*						3	1	1	5
Wild species:									
Francolin, *Francolinus* sp.	5	1			1	4		3	14
cf. Francolin, *Francolinus* sp.					2	5		1	8
Other:									
Aves, bird (indeterminate)	65	11	1	12	3	20	4	16	132
Grand total									201

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